Reg. No. | | | | | | | | | | | [14CE35]

THIRD SEMESTER B.TECH EXAMINATION DECEMBER - 2015

STRENGTH OF MATERIALS

Time: 3 Hours **Maximum Marks:70**

Answer any FIVE full questions choosing at least two from each part.

PART - A

1. a. Define. **04**

i. Young's Modulus

ii. Poisson's ratio

iii. Rigidity Modulus

iv. Bulk Modulus

b. A bar of 8mm×8mm in cross section in subjected to a pull of 7000N. Due to this force, the lateral dimensions of the bar changed to $7.9985 \times 7.9985 mm$. If Rigidity modulus is 80mpa, determine Poisson's ratio and Young's modulus **05**

c. Determine the stresses in various segments of a circular bar shown in fig. Q.1.C. Determine the total change in the length. E=195GPa. **05**

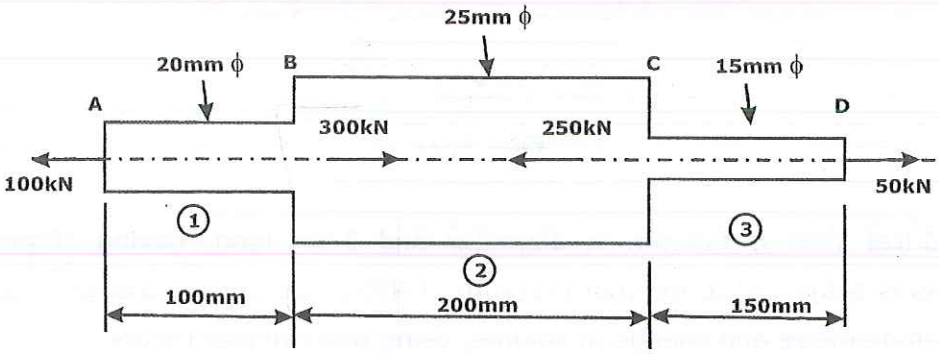

Fig. 1(c)

2. a. A copper bar of 30mm dia is encased centrally in a steel tube of 35mm internal diameter and 40mm external diameter. The ends are rigidly connected. The composite bar is 500mm long and subjected to a pull of 34 KN. Determine the stress in each member and change in length $E_s = 200Gpa \quad E_c = 100Gpa$ **07**

b. A steel bar of 20mm dia pass centrally through a coper tube of 50mm external dia and 40mm internal dia. The tube is closed at ends by tight plates. If temperature of the assembly is raised by 50°c, calculate the stresses developed in copper and steel. $E_s = 200GPa \quad E_c = 100Gpa \quad \alpha = 18 \times 10^{-6} /^0 c \quad \alpha = 12 \times 10^{-6} /^0 c$ **07**

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3. a. Derive the expression for normal and tangential stress σ_n and σ_t on an oblique plane EF shown in Fig Q.3.a, for a general two dimensional stress system. **06**

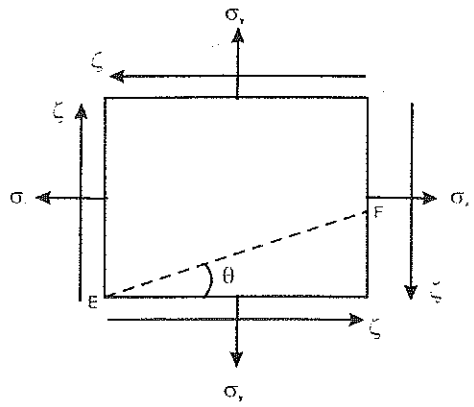

Fig. 3 (a)

b. At a certain point in an elastic material, the state of stress in as shown in Fig. Q.3.b. Determine principal stresses and position of principal planes. **08**

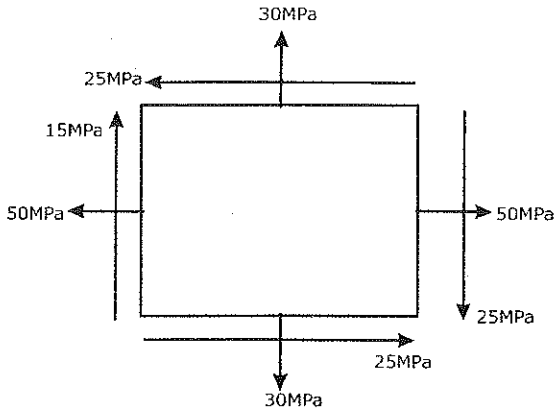

Fig. 3(b)

4. a. A cylindrical shell is 600mm in diameter and 2.5m long, having 10mm wall thickness is subjected to internal pressure of 4MPa. Determine change in length, change in diametre and change in volume, using thin cylinder theory. **07**

b. A thick cylinder of 300mm internal diameter and 80mm well thickness contain a Fluid at a pressure of 6MPa. Determine the maximum and minimum hoop stress across section. Draw hoop stress and redial pressure variation diagram. **07**

5. a. A cantilever of length L in subjected to a UDL of $w/metre$ throughout. Obtain equation for SF and BM and plot SFD, BMD **04**

b. For the beam shown in Fig.Q.5.b. obtain SFD and BMD. Locate point of contraflexure **10**

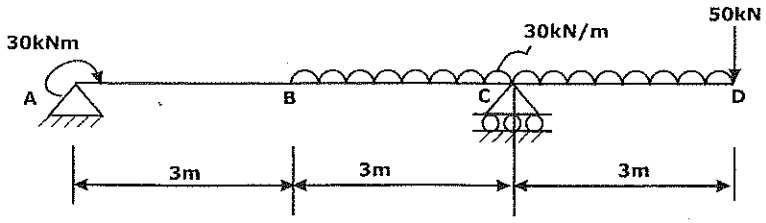

Fig. 5(b)

6. a. Prove with usual notations, the equation of pure bending, $\frac{M}{I}=\frac{\sigma}{y}=\frac{E}{R}$, stating the assumptions. **06**

b. A beam having a square cross section of 20mm×20mm, having a simple supported span of 2m, fails when a point load of 400N is applied at midspan. What UDL per metre length will break a cantilever made of some material, 40mm wide 60mm deep, 3m span **08**

7. a. A hollow circular shaft of 160mm outer dia and 120mm inner dia, transmit 300kW power at 180rpm. Determine **08**

i. Shear stress at inner and outer surface of shaft.

ii. Angle of twist over a length of 6m. $G=84\times10^3\ N/mm^2$.

b. A close coiled vehicle spring is made from a steel wire of 7.5mm dia, having 10 free coils and a mean coil diametre is 80mm. Determine the deflection, when spring is subjected to axial load of 120N. Also calculate stiffness of spring. Given $G=80\times10^3\ N/mm^2$. **06**

8. a. For a beam having rectangular cross section, obtain the equation for shear stress. Prove that the ratio of maximum shear stress to average shear stress is 1.50. **04**

b. A hollow box section 120mm wide, 200mm deep has a uniform wall thickness of 10mm is subjected to a shear force of 120KN. Obtain the equation for shear stress and plot shear stress variation diagram. **10**